Locations

Cathedral Caves – Isle of Eigg
Cupar Silo – Cupar, Fife
Edingham Munitions Factory – Dumfries and Galloway
St Mary's Cathedral – Edinburgh
Tugnet Ice House – Spey Bay, Moray
Wormit Reservoir – Wormit, Fife

Project team

Sam Annand, Craig Gallacher, Paul Gault & Scott Gordon

Profile

This team of artists, musicians and designers are all established in their individual fields, and was initially formed through a strong connection with the east of Scotland, having met at Duncan of Jordanstone College of Art and Design in Dundee. As a group, they've already worked together on projects using a wide range of audio-visual mediums and platforms which fuse a shared identity with the local landscape, architecture, culture and mythology.

Online

soundcloud.com/resonoprogram

Sound

Meyer Sound Leopard soundsystem provided
by Apex Acoustics

Funding

Resono is supported by New Media Scotland's
Alt-w Fund with investment from Creative Scotland.

Programme notes

Saturday 4 June 2016
St Mary's Cathedral

Resono

Project overview

Resono is a sound design project which is taking place in a series of select locations around Scotland. It is based around the concept of playing the echoes and reverberations of a building or space as if it were an instrument.

St Mary's Cathedral

Reverberation time: 6.3 Seconds

Sir George Gilbert Scott designed St. Mary's Cathedral in Edinburgh of 1879. Scott's design for the Cathedral was inspired by the early Gothic churches and abbeys of Scotland with its three spires that stand out on the Edinburgh skyline both by day and by night.

The enormous weight of the 90m central tower (over 5000 tons) is carried on four main pillars and spread through diagonal arches into buttresses in the outer walls, leaving unusually open views inside.

Despite not exhibiting the extremely long reverberation times of the silo in Cupar or reservoir in Wormit, the acoustics in St Mary's Cathedral are rich, refined and very musical. Unlike all of the other spaces we've visited in the project it was designed to exhibit these acoustic properties to enhance and amplify the sound of the choir and pipe organ.

This relatively short but musical reverberation time allowed us to work with rapid, short, percussive sounds that make the room resonate in an almost continuous tone. It also allowed us to experiment with more complex harmonic arrangements that simply couldn't work in other spaces we've visited.

Project details

Traditionally reverberation has mostly been used to amplify sound and music to a large audience, for instance in cathedrals and amphitheatres. However, for *Resono* the intention is to objectify and present the phenomena of reverberation to the audience in a tangible way.

Many of the locations we've chosen for the project have reverberation times in excess of 20–30 seconds which allow us to effectively paint and layer music in the air by using the natural reverb in the spaces to stretch sounds over time.

Most of the sound you're likely to hear in the music is actually resonating in the air around you and reflecting from the walls of the space rather than coming directly from the loudspeakers. The sound from the loudspeakers is effectively 'exciting' the buildings into making music.

To do this we analysed the acoustics of each space to then create a digital simulation of the space in a controlled studio environment elsewhere. This allowed us to work and experiment with the individual characteristics of each location.

We carefully design sounds that would resonate the air inside the structures into making music. The theory is that the music and sounds we've created with the simulation would then have the same effect when amplified into the real structures.

The simulations allow us to get a very close and predictable feel for the how each space reacts to sound but is incomparable to the experience of really being in the space with the air resonating around you.